PHYSICS QUESTION PAPER

Time : 2 Hrs. Max. Marks : 40

Q. 1 Select and write the most appropriate answer from the given alternatives for each sub-question. (8)

(i) According to Kepler's law, the areal velocity of a planet around the sun, always ..... (1)
(a) increases (b) decreases
(c) remains constant (d) first increases and then decreases

(ii) Radius of gyration of a ring about a transverse axis passing through its centre is ..... (1)
(a) 0.5 × diameter of ring (b) diameter of ring
(c) 2 × diameter of ring (d) (diameter of ring)^2

(iii) Poisson's ratio is the ratio of lateral strain to ..... (1)
(a) volume stress (b) shearing strain
(c) longitudinal stress (d) longitudinal strain

(iv) Absorption of water by filter paper is due to ..... (1)
(a) cohesion (b) capillarity
(c) adhesion (d) elasticity

(v) In the equation of a simple harmonic progressive wave of wavelength 'λ' the propagation constant is given by ..... (1)
(a) $\frac{2\pi}{\lambda}$ (b) $\pi\lambda$ (c) $\frac{\pi}{\lambda}$ (d) $\frac{\lambda}{2\pi}$

(vi) The average distance covered by a molecule between two successive collisions is called ..... (1)
(a) free path (b) constant path
(c) mean free path (d) free path per unit time

(vii) An amplitude of a simple pendulum of a period 'T' and length 'L' is increased by 5%. The new period of that pendulum will be ..... (1)
(a) $\frac{T}{8}$ (b) $\frac{T}{4}$ (c) $\frac{T}{2}$ (d) T

(viii) A body cools at the rate of 0.5°C/s when it is at 50°C above the surrounding temperature. The rate of cooling at excess temperature of 30°C over the surrounding temperature is .....(1)
(a) 3° C/s (b) 0.3° C/s
(c) 0.2° C/s (d) 0.1° C/s

Q. 2 (A) Attempt any One : [8]

(i) A torque of 1500 Nm acting on a body produces an angular acceleration of 3.2 rad/s^2. Find M.I. of the body. (2)

(ii) What pressure is required to reduce the volume of a lead blo k by 1% ? (2)
(Given : Bulk modulus of lead = 6×10^9 S.I. unit)

(B) Attempt any Two :

(i) Define 'angle of contact'. State its 'four' characteristics. (3)

(ii) State the points of comparison between progressive waves and stationary waves. (3)

(iii) Assuming the expression for pressure of an ideal gal, show that R.M.S. velocity of gas molecule is directly proportional to the square root of its absolute temperature. (3)

Q. 3 (A) Attempt any One :

(i) Define surface tension. State its S.I. unit and dimensions. (2)

(ii) Represent graphically the variation of potential energy, kinetic energy and total energy of a particle performing S.H.M. with time. (2)

(B) Attempt any Two :

(i) Derive an expression for the kinetic energy of a body rotating with uniform angular speed. (3)

(ii) Prove that work done in stretching the wire is $\frac{1}{2}$ × load × extension, by calculus method. (3)

(iii) State the postulates of Prevost's theory of heat exchanges. Discuss how does it account for the exchange of heat between body and its surroundings. (3)

Q.4 (A) Attempt any Two :

(i) Distinguish between centripetal force and centrifugal force. (2)

(ii) Define molar specific heats of a gas. How are they related to the corresponding principal specific heats ? (2)

(iii) Draw a neat labelled diagram of 'Ritchie's experiment'. (2)

(B) Attempt any One :

(i) Describe how an artificial satellite is launched in an orbit around the earth. Explain the nature of all the possible orbits of a satellite with the help of a suitable diagram. (4)

(ii) Define angular S.H.M. and prove that a bar magnet vibrating in uniform magnetic induction performs uniform angular S.H.M. Obtain an expression for its frequency. (4)

Q.5 Attempt any Two :

(i) Two sound notes have wavelengths $\frac{83}{170}$ m and $\frac{83}{172}$ m in air. These notes when sounded together produce 8 beats per second. Calculate the velocity of sound in air and frequencies of two notes. (4)

(ii) A sonometer wire is in unison with a tuning fork when stretched by a weight of specific gravity 'nine'. On completely immersing the weight in water, wire produces 4 beats per second with the fork. Calculate the frequency of the fork. (4)

(iii) An object of mass 2 kg attached to a wire of length 5 m is revolved in a horizontal circle. If it makes 60 r.p.m. find its (a) angular speed, (b) linear speed, (c) centripetal acceleration, (d) centripetal force. (4)